

MASTERS PROJECT MILESTONE REPORT

Name of the student:

Name of the supervisor(s):

Title of the project (may be tentative): A GLM-Based Penalized Poisson Trick for High-Dimensional Multinomial Classification

Summary: Multinomial mixed models can account for correlation by using group level random effects. However, it has been established that closed-form expressions for the likelihood of multinomial mixed models do not exist regardless of the random effect distribution, except for the special case when there are no covariates.

Previous studies have introduced the “Poisson Trick” and the corresponding Poisson surrogate model to allow the modelling of counts for multiple categories. Nevertheless, existing investigations are typically based on relatively low-dimensional dataset with 2412 observations and only 4 covariates (Yogurt Brand Choice Dataset). As a result, the potential of this approach in more complex, high-dimensional settings remains unexplored.

The “Poisson trick” does not address the statistical challenges associated with high-dimensional data. Rather, it provides a computationally convenient reformulation of the multinomial likelihood. This reformulation yields a more tractable representation that may simplify estimation procedures and create opportunities for extending multinomial models to more complex cases. In particular, it offers a flexible foundation upon which more advanced modelling strategies can be developed when the dimensionality of the data increases.

A natural direction is the high-dimensional extension, where the number of covariates may be large relative to the sample size. In such scenarios, model identifiability, estimation stability, and computational efficiency become increasingly challenging. Based on the tractable likelihood surrogation provided by the “Poisson trick”, this research focuses on incorporating appropriate structural assumptions to enable reliable estimation and interpretation. This aims to leverage the computational advantages of the Poisson reformulation while addressing the statistical complexities inherent in high-dimensional multinomial modelling.

Student’s signature

Supervisor’s signature

Date: